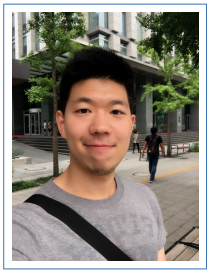
CHANUK YANG

Curriculum Vitae

Xrhuman Laboratory, Center for Artificial Intelligence
KIST(Korea Insitute of Science and Technology
✉ *ych901208@gmail.com*



Research Interest

Robot Navigation, Self-driving car, Visual SLAM, 3D Vision, Robotics

Research Experience

Korea Institute of Science and Technology(KIST)

2025 – Present **Postdoctoral researcher**, *XrHuman Laboratory, Center for Artificial Intelligence*.
- Researching on Artificial General Intelligence, Learning based Visual Odometry using quadruped robot.
- Developed robot sensor platform, including multi cameras, LiDAR and IMU, and performed field tests.
- Developed algorithm of synchronization of multiple cameras.

Advisor : **Dr. Hwasup Lim**, *Head of Center for Artificial Intelligence*, KIST

Hanyang University

2016 – 2025 **Graduate student researcher**, *Machine Monitoring and Control Laboratory*.
- Researched on vehicle localization using various sensors, like GPS, IMU, Camera, and LiDAR
- Developed visual-inertial localization and visual-inertial SLAM and verified both open dataset and experimental vehicle dataset
- Collaborated with Hyundai Motor Company: Led a team developing stuck probability and anti-stuck control algorithm for off-road vehicle
- Collaborated with Hyundai Motor Company: Contributed to driving style prediction project

Advisor : **Prof. Kunsoo Huh**, *Professor, Department of Automotive Engineering*, Hanyang University

Education

2016–2025 : **Ph.D., Automotive Engineering**, *Hanyang University*, Seoul, Republic of Korea.
Thesis title: 6 DOF Ego Vehicle Localization: Filtering and Optimization in Perspective

2009–2016 : **B.S., Mechanical System & Design Engineering**, *Hongik University*, Seoul, Republic of Korea.

Publications

Journal Articles

2024 **Chanuk Yang**, Hayeon O, and Kunsoo Huh. Tightly-coupled, speed-aided monocular visual-inertial localization in topological map. *arXiv preprint arXiv:2411.05497*, 2024.

2024 **Chanuk Yang** and Kunsoo Huh. Comparision of loosely and tightly coupled visual inertial odometry using lidar prior map. *Transactions of the Korean Society of Automotive Engineers,*, 2024.

2024 **Chanuk Yang** and Kunsoo Huh. 6 dof vehicle pose estimation considering lidar odometry initial condition. *IEEE Access*. IEEE, 2024.

2021 Dongchan Kim, Hyukju Shon, Nahyun Kweon, Seungwon Choi, **Chanuk Yang**, and Kunsoo Huh. Driving style-based conditional variational autoencoder for prediction of ego vehicle trajectory. *IEEE Access*, volume 9, pages 169348–169356. IEEE, 2021.

In Conference Proceedings

2024 Park Sungjun, **Chanuk Yang**, Han Sangwon, and Kunsoo Huh. Robust mpc based lateral control system design for autonomous vehicle considering localization uncertainty. In *2024 IEEE International Intelligent Transportation Systems Conference (ITSC)*. IEEE, 2024.

2024 Hayeon O, **Chanuk Yang**, and Kunsoo Huh. Sesame: Simple, easy 3d object detection with point-wise semantics. In *Computer Vision – ACCV 2024*, 2024.

2024 Junhan Kang, Um Jin, Kunsoo Huh, **Chanuk Yang**, et al. A study on estimation of stuck probability in off-road based on ai. Technical report, SAE Technical Paper, 2024.

2021 Seungwon Choi, Nahyun Kweon, **Chanuk Yang**, Dongchan Kim, Hyukju Shon, Jaewoong Choi, and Kunsoo Huh. Dsa-gan: driving style attention generative adversarial network for vehicle trajectory prediction. In *2021 IEEE International Intelligent Transportation Systems Conference (ITSC)*, pages 1515–1520. IEEE, 2021.

Awards

2021 - 2023 **Selected on outstanding project in Hyundai Motor Company**.
Project Title: Deep learning based stuck probability and anti-stuck control algorithm for off-road vehicle

2025 **Outstanding Researcher Award in AI-Robotics Research Center, KIST** .

Computer skills

Programming Languages	C, C++, Python
Tools	ROS1, ROS2, Eigen, Ceres, GTSAM, Pytorch, Docker
Simulator	Carsim, Carmaker

Courseworks (Selected)

Hanyang Univ.	Optimization, Software Engineering, Computer Architecture, Computer Algorithm, Linear system, Intelligent Vehicle Recognition System, Vehicle Dynamics&Control
Online	Sensor Fusion(Certificated Nanodegree in Udacity)

Language

Korean(Native), English(Fluent)

Referees

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